
A Short Linear Algebra Primer

eigenvalues & eigenvectors, definite & semidefinite matrices, Cholesky

Why we need this

Bayesian linear regression (and Gaussians in general) reduce to a few matrix operations on *symmetric* matrices — covariances Σ and precisions Σ^{-1} :

- The posterior needs us to invert matrices

$$p(\mathbf{w} \mid \mathcal{D}) = \mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}_N, \boldsymbol{\Sigma}_N), \quad \boldsymbol{\Sigma}_N = (\lambda \mathbf{I}_D + \beta \mathbf{X}^\top \mathbf{X})^{-1}, \quad \boldsymbol{\mu}_N = \boldsymbol{\Sigma}_N \beta \mathbf{X}^\top \mathbf{y}.$$

and to obtain $\boldsymbol{\Sigma}_N$.

- Every covariance matrix must be a **valid** covariance — this is where *positive (semi)definiteness* enters.
- **Eigenvalues/eigenvectors** describe the shape of a Gaussian (the “uncertainty ellipse”) and tell us when a matrix is (nearly) singular (ill-conditioning).
- **Cholesky** is the fast, stable workhorse for solving these systems, computing $\log \det$, and sampling from a Gaussian.

This primer covers exactly those pieces.

Eigenvalues and eigenvectors

Definition

A nonzero vector v is an **eigenvector** of a square matrix A with **eigenvalue** λ if

$$Av = \lambda v.$$

A merely *stretches* v by the factor λ (direction unchanged).

How to find them (2×2): solve $\det(A - \lambda I) = 0$ for λ , then $(A - \lambda I)v = 0$ for v .

Meaning: eigenvectors are the “natural axes” of the transformation; eigenvalues say how much it scales along each axis. $\lambda = 0$ means A collapses that direction $\Rightarrow A$ is *singular* (not invertible).

Watch [Eigenvectors and eigenvalues](#) | Chapter 14, Essence of linear algebra

Worked example

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}: \det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0$$

$$\Rightarrow \lambda = 3, 1.$$

$$\lambda=3: v \propto (1, 1); \quad \lambda=1: v \propto (1, -1).$$

Symmetric matrices and the covariance ellipse

Spectral theorem (symmetric $A = A^T$)

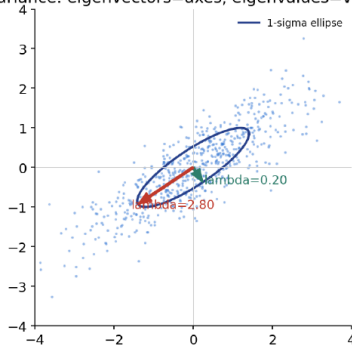
A real symmetric matrix has **real** eigenvalues and **orthonormal** eigenvectors, and factors as $A = Q\Lambda Q^T$ with Q orthogonal ($Q^T Q = I$) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Covariances are symmetric, so this always applies. For a Gaussian with covariance Σ :

- the **eigenvectors** are the axes of the uncertainty ellipse;
- the **eigenvalues** are the variances along those axes (axis lengths $\propto \sqrt{\lambda}$).

A large spread of eigenvalues (large *condition number* $\lambda_{\max}/\lambda_{\min}$) means a stretched, nearly-degenerate ellipse — the ill-conditioning we saw in ridge regression.

Covariance: eigenvectors=axes, eigenvalues=variances



Eigenvectors and eigenvalues of a Gaussian covariance

$$\mathbf{z} = \begin{bmatrix} x \\ y \end{bmatrix} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Unit eigenvectors and their eigenvalues

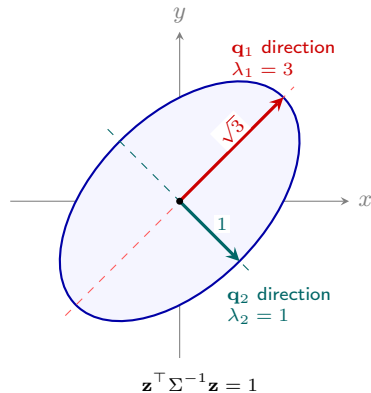
$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \lambda_1 = 3, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \lambda_2 = 1.$$

Verification: $\Sigma \mathbf{q}_i = \lambda_i \mathbf{q}_i$

$$\Sigma \mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \mathbf{q}_1,$$

$$\Sigma \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \mathbf{q}_2.$$

$$\Sigma = Q \Lambda Q^\top, \quad Q = [\mathbf{q}_1 \ \mathbf{q}_2], \quad \Lambda = \text{diag}(3, 1).$$



Arrows show $\sqrt{\lambda_i} \mathbf{q}_i$, the ellipse semiaxes.
Enclosed probability: approximately 39%.

Eigenvectors give axis directions. Eigenvalues give variances along those axes.

Eigenvectors and coordinates along principal axes

The eigenvectors are fixed unit vectors that define directions:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

For a point $\mathbf{z} = [x, y]^\top$, its coordinates along these directions are scalars:

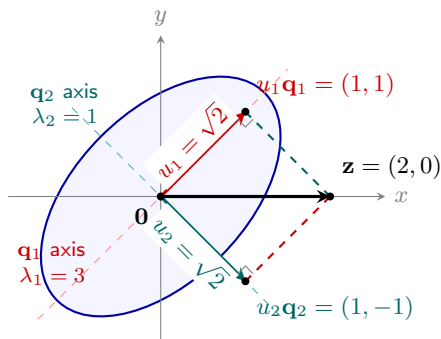
$$u_1 = \mathbf{q}_1^\top \mathbf{z} = \frac{x+y}{\sqrt{2}}, \quad u_2 = \mathbf{q}_2^\top \mathbf{z} = \frac{x-y}{\sqrt{2}}.$$

Example: If $\mathbf{z} = [2, 0]^\top$, then $u_1 = u_2 = \sqrt{2}$. Reconstructing the point gives

$$\mathbf{z} = u_1 \mathbf{q}_1 + u_2 \mathbf{q}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

For $\mathbf{z} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\right)$,

$u_1 \sim \mathcal{N}(0, 3)$, $u_2 \sim \mathcal{N}(0, 1)$, u_1 and u_2 are independent.



$$\text{Ellipse: } \mathbf{z}^\top \Sigma^{-1} \mathbf{z} = 1, \quad \Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

For a nonzero mean, use $u_i = \mathbf{q}_i^\top (\mathbf{z} - \boldsymbol{\mu})$.

Positive-definite and positive-semidefinite matrices

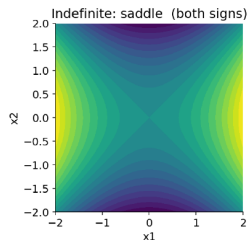
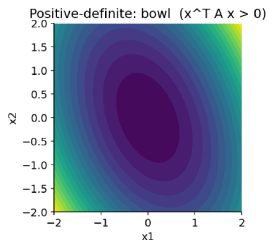
Definition (symmetric A)

A is **positive-definite** (PD) if $\mathbf{x}^\top A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$; **positive-semidefinite** (PSD) if $\mathbf{x}^\top A \mathbf{x} \geq 0$.
Equivalently: **PD** $\Leftrightarrow$ all eigenvalues > 0 ; **PSD** $\Leftrightarrow$ all eigenvalues ≥ 0 .

Examples.

- $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$: eig 2, 1 $> 0 \Rightarrow$ **PD**.
- $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: eig 2, 0 $\Rightarrow$ **PSD** (singular).
- $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$: eig 1, -1 $\Rightarrow$ *indefinite* (saddle).

Where it is needed: a covariance must be **PSD** (variances ≥ 0); a valid, non-degenerate Gaussian needs a **PD** covariance (so Σ^{-1} and $\log \det \Sigma$ exist); the BLR precision $\lambda \mathbf{I} + \beta \Phi^\top \Phi$ is **PD** for $\lambda > 0$; convex optimisation needs a PD Hessian; and **Cholesky exists iff the matrix is PD**.



$\mathbf{x}^\top A \mathbf{x}$: a PD matrix gives a *bowl* (unique minimum); an indefinite matrix gives a *saddle*.

Cholesky decomposition — what it is

Definition

Every **positive-definite** symmetric A factors *uniquely* as

$$A = LL^{\top},$$

where L is **lower-triangular** with positive diagonal. L is a “matrix square root” of A ; the factorization *exists iff* A is *PD* (so it doubles as a PD test).

Worked example

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{pmatrix} = LL^{\top}.$$

Check: $L_{11} = \sqrt{4} = 2$, $L_{21} = 2/2 = 1$,
 $L_{22} = \sqrt{3 - 1^2} = \sqrt{2}.$

Cost $\approx \frac{1}{3}n^3$ flops — about *half* the work of an LU factorization, and numerically very stable for PD matrices (no pivoting needed). In code: $L =$

```
scipy.linalg.cho_factor(A)
```

```
x = scipy.linalg.cho_solve(L, b) # solve  $Ax = b$ 
```

Cholesky — when and why we use it

Given a PD matrix $A = LL^\top$ (with L triangular), everything below is a cheap *triangular solve*:

- **Solve a linear system** $Ax = b$: forward-solve $Lz = b$, then back-solve $L^\top x = z$. This is how we get the posterior mean μ_N — *never* form A^{-1} .
- **Log-determinant** $\log \det A = 2 \sum_i \log L_{ii}$ — stable and free (needed for the NLPD and the evidence).
- **Sample from a Gaussian** $\mathcal{N}(\mu, \Sigma)$: draw $z \sim \mathcal{N}(0, I)$, return $\mu + Lz$ where $\Sigma = LL^\top$ (L correlates the white noise).
- **Invert an SPD matrix / get Σ_N** : solve $AX = I$ column by column.

Why not just use `inv`?

Explicitly inverting is slower and numerically worse: it roughly *squares* the condition number and amplifies round-off when features are correlated. Cholesky solves are the stable, standard choice — and they fail exactly when A is not PD, which is a useful warning.

In this course: the BLR posterior uses $A = \lambda I + \beta \Phi^\top \Phi$ (PD), so `cho_factor/cho_solve` give μ_N , Σ_N , and $\log \det A$ for the evidence.

Summary & references

- **Eigen-decomposition** $A = Q\Lambda Q^\top$ describes a symmetric matrix's axes (Q) and scalings (Λ); for a covariance these are the uncertainty-ellipse axes and variances.
- **PD** ($\lambda_i > 0$) / **PSD** ($\lambda_i \geq 0$): covariances are PSD, valid Gaussians and BLR precisions are PD; PD is what makes " $x^\top Ax$ " a bowl and Cholesky possible.
- **Cholesky** $A = LL^\top$ (PD only) is the fast, stable tool for solving systems, log-determinants, and Gaussian sampling — use it instead of `inv`.

References:

G. Strang, *Introduction to Linear Algebra*;

G. Golub & C. Van Loan, *Matrix Computations* (Cholesky, conditioning);

L. Trefethen & D. Bau, *Numerical Linear Algebra*;

S. Boyd & L. Vandenberghe, *Convex Optimization*, App. A (PD matrices);

Petersen & Pedersen, *The Matrix Cookbook*.

This document was developed and converted into LaTeX slides and formatted with assistance from ChatGPT (OpenAI) and CoPilot (Microsoft). The instructor modified the source text and verified the final structure and wording.